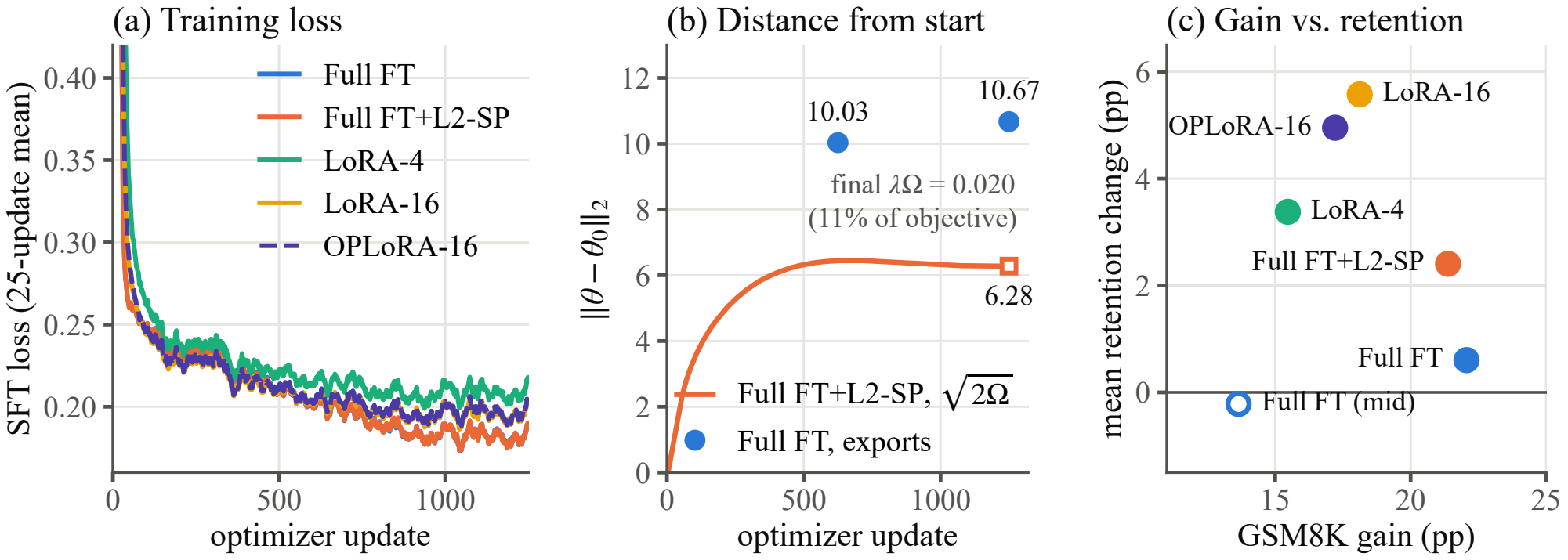

(a) Training loss
SFT loss (25-update mean)
optimizer update
Full FT
Full FT+L2-SP
LoRA-4
LoRA-16
OPLoRA-16
0.40
0.35
0.30
0.25
0.20
0
500
1000
(b) Distance from start
$\|\theta - \theta_0\|_2$
optimizer update
10.03
10.67
final $\lambda\Omega$ = 0.020
(11% of objective)
6.28
Full FT+L2-SP, $\sqrt{2\Omega}$
Full FT, exports
12
10
8
6
4
2
0
(c) Gain vs. retention
mean retention change (pp)
GSM8K gain (pp)
LoRA-16
OPLoRA-16
LoRA-4
Full FT+L2-SP
Full FT
Full FT (mid)
6
4
2
0
15
20
25